

Assessment of Radiation Doses for Paediatric Head, Chest, and Abdominopelvic Computed Tomography Examinations in a Teaching Hospital in Ghana

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Abstract

Introduction: With the increasing number of pediatric computed tomography (CT) examinations, there is a need to optimise protocols for children by adopting examination-specific protocols customised to the patient's age, size, imaging region, and clinical indication. This study aimed to assess the radiation doses in pediatric CT examinations and compare them to international standards.

Material and methods: A cross-sectional retrospective study design was adopted to probe patient records at the radiology department of a teaching hospital in Ghana. Thus, scan parameters, volume computed tomography dose index (CTDI_{vol}), dose length product (DLP), as well as demographic data, were recorded from 496 pediatric patients (age 0-15 years) undergoing head, chest, and abdominopelvic CT examinations. Local Diagnostic Reference Levels (LDRLs) were established using the 75th percentile of patient dose values for each protocol and age group. These local levels were then compared with DRLs from other studies.

Results: Head CT was the most performed examination (35.0%) compared to chest (32.0%) and abdominopelvic (33.0%). The male group recorded the highest (59.1%) percentage of CT examinations compared to the female group. While LDRL values from this study were generally lower than data from other studies, the CTDI_{vol} and DLP for head scans of patients between 11 and 15 years were found to be higher than the data from other studies.

Conclusions: Our study has established LDRLs for standard pediatric CT examinations in the teaching hospital. The LDRLs were generally lower than those reported in other studies, except for head scans in patients aged 11 to 15 years. These findings suggest that there are opportunities for further optimisation of pediatric CT imaging protocols at this facility.

Keywords: Diagnostic reference levels (DRLs), Computed Tomography (CT), Computed Tomography Dose Index (CTDI_{vol}), Dose Length Product (DLP), Teaching Hospital

Introduction

Computed tomography (CT) is a valuable diagnostic tool. From the time of its introduction to clinical use, it has seen rapid advances and developments, resulting in a wide range of applications¹. However, CT examinations contribute significantly to the collective diagnostic radiation dose; about 41-75% of the overall medical dose². The increased number of CT examinations and relatively high individual doses have

raised concerns about potential health risks²⁻⁴. In the United States, over 62 million CT scans are obtained annually, including more than 4 million for children^{5,6}.

In recent years, radiological imaging of children has proven to be among the fastest-growing fields, even though pediatric CT may result in significantly higher lifetime radiation risk than adult CT⁷. With the increasing number of pediatric CT examinations, there is a need to optimise protocols for children by adopting examination-specific protocols customised

to the patient's age, size, imaging region, and clinical indication³. This is to ensure that each patient's dose is as low as reasonably achievable (ALARA). Diagnostic Reference Levels (DRLs), introduced by the International Commission on Radiological Protection (ICRP), guide in achieving radiation protection for patients. DRLs for CT examinations are important for dose optimisation and vary based on facility/local, regional or national data. In addition to serving as an indicator of typical radiological practice within a facility, region, or country, DRLs are used to identify abnormal radiation doses for standard diagnostic medical X-ray imaging^{8,9}.

In Ghana, there is currently no pediatric regulatory DRL against which dose parameters could be compared to improve efficient dose monitoring and optimisation, although there are a considerable number of installed CT scanners¹⁰. Establishing National DRLs (NDRLs) based on large-scale surveys of median doses indicating usual practice for different patient groups can help identify typical radiological practices and optimise the protocols. However, several countries have no dedicated pediatric protocols. Thus, adult protocols are used for children, resulting in higher radiation doses^{11,12}. Hence, this study aimed to estimate the representative radiation doses of pediatric patients undergoing CT examinations at a teaching hospital in Ghana and compare the findings to international standards. Thus, scan parameters, volume computed tomography dose index (CTDI_{vol}), dose length product (DLP) and effective dose (ED) data for three pediatric CT examinations (head, chest, abdominopelvic) were compared with international standards to evaluate local practices. To the best of our knowledge, this study is the first of its kind in the country to develop age-based local pediatric DRLs for a facility.

Materials and Methods

Patients

A cross-sectional retrospective study design was employed. From the institution's Picture Archiving and Communication System (PACS), an audit of all paediatric CT scans performed from January 2019 to December 2021 at a teaching hospital in Ghana was conducted. The targeted population was paediatric patients aged 0-15 years who had undergone CT examinations at the Radiology Department. This facility was selected because it records a relatively higher number of pediatric cases, providing a sufficiently large population for the study. Purposive sampling was used to gather data from 496 paediatric patients undergoing head, chest, and abdominopelvic CT examinations. Information about the scan parameters, CTDI_{vol}, and DLP values was recorded for all patients. Radiation doses for each body region were compared with DRLs from other stud-

ies. Body weight was not available for all records; thus, patients were grouped by age for analysis.

CT Acquisition Parameters

All CT scans were performed using a Toshiba Aquilion One 640-slice multi-slice scanner installed in December 2012. The scanner uses a 320-detector row configuration and operates with a scan speed of 0.275 seconds. The Adaptive Iterative Dose Reduction 3-Dimensional (AIDR 3D) algorithm was used for image reconstruction. The specific scan parameters applied for each CT protocol and age group (including tube potential (kVp), milliamper second (mAs), slice thickness, and scan length) are summarised in **Table 2**. For each patient, the CTDI_{vol} and DLP values were automatically recorded by the scanner and extracted from the dose reports. The CTDI_{vol}

Table 1: Scanner specifications and range of acquisition parameters for paediatrics.

Manufacturer	Toshiba
Model	Aquilion One Vision 640-slice scanner
Year of Installation	December 2012
Scan speed	0.275 seconds
Number of Detectors	320-detector row
Iterative reconstruction algorithm	AIDR 3-D

Table 2. Scan parameters per age group for the various CT examinations: Scan length and mAs are represented as mean $\pm$ standard deviation (range). Tube potential and slice thickness are presented in value (percentage of cases).

Scan Parameters Per Examination						
Examination	Age Groups (years)	N	Tube Potential (kVp)	Milliamper second (mAs)	Slice Thickness (mm)	Scan Length (mm)
Head	< 1	-	-	-	-	-
	1 – 5	66	100 (80%), 120 (20%)	164 $\pm$ 77.4 (37 – 300)	5.0 (100%)	163 $\pm$ 19.6 (95 – 215)
	6 – 10	49	100 (63%), 120 (37%)	191 $\pm$ 57.4 (37 – 225)	5.0 (100%)	181 $\pm$ 30.9 (135 – 230)
	11 – 15	56	100 (14%), 120 (86%)	215 $\pm$ 29.2 (121 – 225)	5.0 (100%)	177 $\pm$ 33.5 (135 – 240)
	< 1	-	-	-	-	-
	1 – 5	22	80 (27%), 100 (73%)	29 $\pm$ 6.9 (25 – 45)	2.0 (18%), 3.0 (32%), 5.0 (50%)	174.8 $\pm$ 36.9 (130 – 213)
Chest	6 – 10	40	100 (60%), 120 (40%)	35 $\pm$ 5.7 (25 – 45)	3.0 (42%), 5.0 (58%)	282 $\pm$ 39.1 (204 – 335)
	11 – 15	98	100 (67%), 120 (33%)	40.4 $\pm$ 6.7 (25 – 60)	1.0 (2%), 3.0 (10%), 5.0 (88%)	303 $\pm$ 56.3 (215 – 420)
	< 1	-	-	-	-	-
	1 – 5	36	80 (25%), 100 (75%)	31 $\pm$ 5.5 (24 – 40)	3.0 (58%), 5.0 (42%)	328 $\pm$ 26.9 (285 – 415)
Abdominopelvic	6 – 10	45	100 (82%), 120 (18%)	42 $\pm$ 29.7 (25 – 225)	3.0 (13%), 5.0 (87%)	401 $\pm$ 47.1 (310 – 465)
	11 – 15	84	100 (49%), 120 (51%)	31 $\pm$ 5.5 (24 – 40)	5.0 (100%)	328 $\pm$ 26.9 (285 – 415)

values were based on a 16 cm-diameter phantom for head examinations and a 32 cm-diameter phantom for chest and abdominopelvic examinations, in accordance with standard manufacturer configuration. For contrast-enhanced examinations involving the trunk, both single-phase and multi-phase protocols were used. In all cases, the reported DLP values represent the cumulative dose for the entire examination, including all scan phases, consistent with ICRP Publication 135 recommendations.

Inclusion and Exclusion Criteria

Contrast-enhanced and non-contrast-enhanced examinations performed for clinical indications; trauma, seizures, pneumonia, pleural effusion, mass/tumour (for both chest and abdominopelvic examinations), hydronephrosis, and abdominal pain were included in the study. All head scan indications were single-phase and non-contrast; however, chest and abdominopelvic scans included both single and multi-phase (pre-contrast and post-contrast administration) examinations. Apart from abdominopelvic examinations, data from all multi-region CT examinations (e.g., head-and-neck, chest+abdomen+pelvis) were excluded from the study. For each age group, a minimum of 20 patient records containing complete dosimetric information, including age, scan parameters, $CTDI_{vol}$ and DLP, were included in the analysis. Data gathered for the various indications and examinations for the age group <1 year was excluded because they were not up to the minimum sample of 20 patients' dosimetric data required for the study. Additionally, examinations with incomplete information (patient age, date of examination, dose indicators, clinical history, and parameters) were excluded from the study.

Data Analysis

All analyses were performed using Microsoft Excel LTSC 2021 version and SPSS version 25.0. Shapiro-Wilk normality test of the distribution of dosimetric variables was assessed, and the results indicated a non-symmetrical distribution ($p < 0.05$). Consequently, descriptive statistics are presented as median and interquartile range (IQR), which are more appropriate for non-parametric data. The ED for each examination was estimated by multiplying the DLP by the conversion factor for k, as adopted from¹³. For each protocol, the median, 75th percentile, range and IQR values for $CTDI_{vol}$, DLP, and ED were calculated. The 75th percentiles of the radiation dose distribution, as calculated using $CTDI_{vol}$ and DLP, were used to determine the LDRLs. $CTDI_{vol}$ and DLP values were compared with pediatric DRLs from the United States of America (USA), China, Jordan, and South Africa reported by Kanal et al.; Kang et al.; Rawashdeh et al.; and van der Merwe & Mahomed¹⁴⁻¹⁷ respectively.

Ethics

Before the study, ethical clearance with a relinquishment of patient informed consent was sought and approved by the Research Office of the teaching hospital. Further approval to access the patient records was obtained from the head of the Radiology Department. Since the preservation of participants' rights and well-being is a major concern of scientific ethics, confidentiality and anonymity were ensured for all the patients' data used.

Results and Discussion

Demographic Statistical Analysis

Figures 1a and 1b depict the frequency of CT examinations for the various body regions and the distribution of clinical indications for the various examinations, respectively. As illustrated in Figures 1a and 1b, head CT emerged as the most frequently requested examination. A greater proportion of

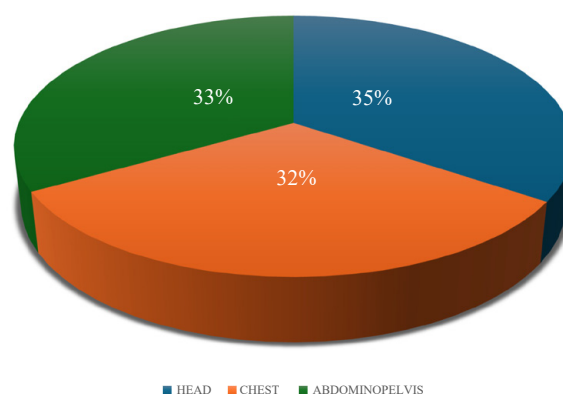


Figure 1a. Frequency of CT examinations performed for the various body regions.

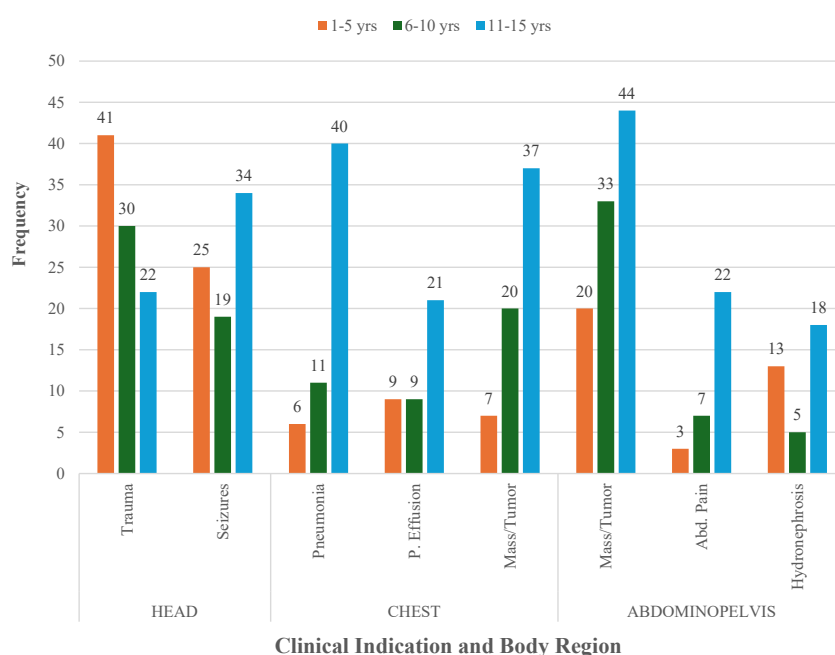


Figure 1b. Frequency of the CT examinations performed for the various clinical indications and age groups.

chest and abdominopelvic scans were performed in older children (11-15 years), reflecting a possible age-related pattern in clinical indications and imaging preferences. The male group recorded the highest (59.1%) percentage of CT examinations compared to the female group. Data for less than 1 year was not up to the minimum number of cases required, suggesting the use of other modalities for that age group.

Assessment of radiation doses in pediatric CT examinations reveals a notable preference for head scans, accounting for 35% of the procedures. It may also indicate that other modalities were used for chest and abdomen examinations. This trend aligns with international studies such as those by Gao et al,¹⁸ and Kanel et al,¹⁴ underscoring the global focus on head trauma and neurological conditions in this demographic. However, our findings diverge from Endalamaw et al.,¹⁹ regarding the epidemiology of childhood cancer, suggesting potential regional differences in disease presentation or healthcare practices. The higher incidence of mass/cancer indications in the 11 - 15 age group (**Figure 1a**) may reflect a delayed presentation or diagnosis in the Ghanaian context, highlighting the need for improved early detection and reporting mechanisms. This could also mean increased use of non-ionising radiation for patients below 11 years. Furthermore, the gender disparity observed, with males undergoing more CT examinations, prompts a discussion on gender-specific health issues and access to care. These insights are crucial for tailoring healthcare policies and optimising radiation safety protocols to the specific needs of the pediatric population in Ghana.

Local Diagnostic Reference Levels

The mean scan length and mAs values were evaluated for each age group, while the kVp and slice thickness were reported as discrete values with their corresponding frequencies as shown in **Table 2**. Our analysis of these exposure parameters indicates adherence to the recommended DRLs set by the Paediatric EU Standards and corroborates the findings of Agorsah²⁰ and Al-Rammah²¹. Automatically generated CTDI_{vol} and DLP from the dose report were employed as pivotal dose indicators, reflecting the radiation exposure during pediatric CT examinations. **Figure 2** (labelled A to F) shows the respective median and 75th percentile CTDI_{vol} and DLP values of the CT examinations for the various pediatric age groups. Notably, both indicators exhibited an age-dependent increase for all protocols. This shows that a one-size-fits-all approach is consciously avoided to ensure dose optimisation across different age groups in the teaching hospital. For all age groups, large variations were seen in DLP values for the head scans than for the chest and abdominopelvic regions (**Figure 2D-F**). Large variations were also observed between DLPs in each patient group for all protocols. The use of variable scan length for the same patient group could account for these wide-range values. This might be due to differences in the radiographers' scan techniques²². Other researchers have observed variations in radiation doses for the same patients for small differences in scan techniques²³.

Based on the 75th percentile values of the CTDI_{vol} and DLP, LDRLs were established for each CT protocol and age group. These values are summarised in **Figure 2**. For head CT, the

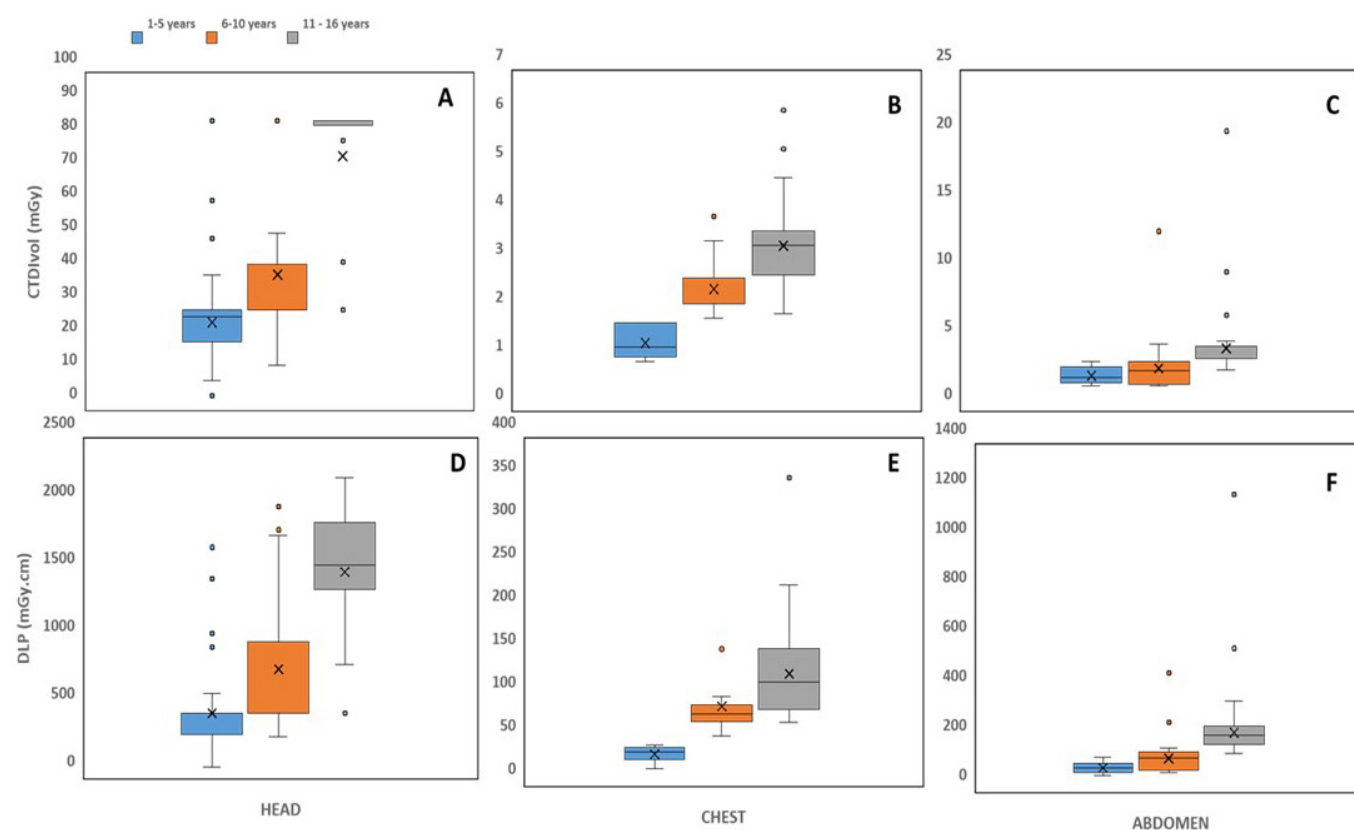


Figure 2. Boxplots for CTDI_{vol} (A, B, C) and DLP (D, E, F) values for head, chest and abdominopelvic examination. The x indicates mean values, and the midline in the box indicates median values; the whiskers (T bars) that extend from the box represent minimum and maximum values (without outliers). The outliers are the points beyond the whiskers, while the box represents the 25th-75th percentile.

75th percentile CTDI_{vol} values were 29.6 mGy for children aged 1-5 years, 43.8 mGy for 6-10 years, and 86.0 mGy for 11-15 years, with corresponding DLP values of 473.7, 1025.0, and 1925.4 mGy-cm, respectively. For chest CT, the 75th percentile CTDI_{vol} values were 1.8, 2.4, and 3.7 mGy for the 1-5, 6-10, and 11-15-year age groups, respectively, with corresponding DLP values of 43.2, 85.8, and 157.2 mGy-cm. In abdominopelvic CT, the 75th percentile CTDI_{vol} values were 3.2 mGy for 1-5 years, 3.6 mGy for 6-10 years, and 4.7 mGy for 11-15 years, with respective DLP values of 111.4, 156.4, and 261.9 mGy-cm. These LDRL values serve as local benchmarks for paediatric CT dose optimisation in the facility and can guide future dose audits and protocol refinement efforts.

DRL Comparison to other studies

In line with ICRP Publication 135, the LDRLs were established as the 75th percentile values of the distribution of individual patient dose metrics (CTDI_{vol} and DLP) within a single health-care facility. This approach reflects typical local practice and is appropriate for setting facility-based DRLs. The 75th percentile of CTDI_{vol} and DLP of the various age groups from this study was compared to DRL values from Jordan, the USA, China and South African national surveys¹⁴⁻¹⁷ as represented in **Table 3**. CTDI_{vol} values for head scans recorded in this study were found to be equal to or lower than the values reported by the comparing studies, except for values recorded by van der Merwe & Mahomed, (2020)¹⁷: (29.6 vs 20.3 mGy) for 1-5 years, (43.8 vs 20.33 mGy) for 6 - 10 years and 11-15 years, (86.0 vs 32.14 mGy). Although this study's CTDI_{vol} values for head examinations were generally lower than those reported by Kanal et al.¹⁴ and Kang et al.¹⁵, the DLP values for the 6-10-year age group were found to be higher. This discrepancy likely reflects

the use of longer scan lengths, potentially due to the inclusion of 'safety margins' by radiographers. While such caution is understandable, it may inadvertently lead to increased radiation exposure. Furthermore, this study's CTDI_{vol} and DLP values for head scans in the 11-15-year age group were higher than those reported in all comparative studies, indicating a need to review the scanning protocol for this age group.

For chest and abdominopelvic examinations, the CTDI_{vol} and DLP values recorded in this study were generally lower than those reported in the comparative literature, despite the inclusion of cumulative DLP from multi-phase examinations. Additionally, significant variations were observed among the data reported from Jordan, the USA, China, and South Africa, highlighting the need for a critical evaluation of scanning protocols. It is worth noting that variations in methodology, such as whether the reported DLP values represent cumulative exposures, the type of examination (single-phase vs. multi-phase), the clinical indications included, and the phantom size used to estimate CTDI_{vol}, can all influence comparisons. These methodological differences must be considered when interpreting dose data and making direct comparisons between DRL values across studies.

The broad range of DLP values observed within patient groups further underscores the influence of radiographer technique on dose variability. This highlights the critical need for standardized training and strict adherence to imaging protocols to minimize dose discrepancies and enhance patient safety. By addressing these discrepancies and reviewing protocols, particularly for head CT examinations in older children, radiation doses might be reduced and the overall safety and effectiveness of pediatric CT imaging practices within the facility will be improved.

Table 3. Comparison of CT DRLs from this study with international pediatric CT DRLs.

Age Groups	CTDI _{vol} (75th Percentile)					DLP (75 th Percentile)				
	This Study	(Rawashdeh et al., 2019)	(Kanal et al., 2022)	(Kang et al., 2021)	(van der Merwe & Mahomed, 2020)	This Study	(Rawashdeh et al., 2019)	(Kanal et al., 2022)	(Kang et al., 2021)	(van der Merwe & Mahomed, 2020)
HEAD										
< 1 year	-	47.8	23	36.5	20.3	-	743.7	344	488	311.30
1-5 years	29.6	54.7	31	43.4	20.3	473.7	981.8	518	635	362.40
6-10 years	43.8	65.0	46	49.1	22.33	1025.0	1129.5	910	722.5	457.20
11-15 year	86.0	60.7	46	51.1	32.14	1925.4	1207.9	910	852	746.10
CHEST										
< 1 year	-	5.6	1.6	6.1	5.66	-	124	31	108.3	153.50
1-5 years	1.8	7.4	2.4	6.4	3.27	43.2	222.1	58	136.3	105.10
6-10 years	2.4	12.9	2.9	6.8	4.73	85.8	416.4	95	186.7	136.40
11-15 years	3.7	12.9	7.2	9.6	6.97	157.2	496.4	272	292.6	325.10
ABDOMINOPELVIC										
< 1 year	-	12.6	2.4	10.5	6.17	-	325.1	60	251	191.50
1-5 years	3.2	19.8	2.9	10.3	4.73	111.4	408.7	100	284	187.80
6-10 years	3.6	12.8	4.6	11.4	4.73	156.4	460.5	170	353.2	203.30
11-15 years	4.7	16.1	7.9	13.0	8.40	261.9	807	358	523	371.10

Effective Dose Estimation

The concept of effective dose is central to our understanding of radiation risk, particularly in pediatric patients²⁴. It is a measure of overall radiation detriment, and it is used in planning and optimising radiation exposures because it is intended to provide a measure of cumulative radiation detriment resulting from stochastic effects^{25,26}. By employing the conversion factors from ICRP publication 103, our study provides a nuanced estimation of effective doses across various age groups and body regions. The results in **Table 4** reveal that despite large variations in DLP values across age groups, the corresponding effective doses showed less dramatic variation. As an example, median DLP for chest examinations for 11-15 years is 119.4, and for 1-5 years is 38.4 (ratio of 3.1:1). However, the ratio of effective doses is two times lower ($2.82/1.85 = 1.5:1$). Also, the age-related increase in effective dose for head and chest regions, with a notable peak in the 11-15 age group, reflects the changing radiation sensitivity and body composition of growing children. These findings highlight the necessity of age-specific protocols that account for the developmental stage of pediatric patients. Additionally, the effective dose is highest in the abdominopelvic region, compared with the head and chest. This is due to differences in anatomical locations, organ sensitivity and scan parameters. Thus, abdominal CT scans expose more tissues over a longer range, with more radiosensitive organs and higher technical settings.

Study Limitations

According to ICRP Publication 135, weight-based grouping is the preferred method for determining DRLs in pediatric trunk examinations. However, due to the absence of weight data in the patient records, age-based groupings were used instead. While this approach is less optimal, it still offers meaningful insights, particularly for head CT examinations where age bands are recommended. This limitation highlights the need for a systematic collection of weight information in future audits. Comparison of DRLs with international standards was also limited due to the disparity in age stratification for DRL data. Finally, enough data were not available to report on representative doses for the age group <1 year. An image quality assessment was not performed in this study. However, the data was collected retrospectively from examinations reported by radiologists; hence, it was assumed that all examinations were of diagnostic quality.

Conclusion

This study estimated 75th percentile CT dose indicators like $CTDI_{vol}$ and DLP for 3 pediatric CT examinations and compared them with DRLs in other studies. The results showed

Table 4. Median DLP and effective dose estimates for this study using conversion factor k, from ICRP publication 103 (Deak et al., 2010).

DLP (mGy. cm)							Conversion Factors k , (mGy ⁻¹ .cm ⁻¹) ¹³	Effective Dose (mSv)
Age Groups	N	Median	IQR	Min.	Max.			
HEAD								
< 1 year		-	-	-	-	0.0087	-	
1-5 years	66	473.7	148.0	79.0	1699.6	0.0054	2.5579	
6-10 years	49	473.7	551.3	298.7	2000.7	0.0035	1.6579	
11-15 years	56	1570.6	526.85	473.7	2215.7	0.0027	4.2406	
CHEST								
< 1 year		-	-	-	-	0.0739	-	
1-5 years	22	38.4	13.8	18.6	45.8	0.0482	1.8508	
6-10 years	40	81.8	12.6	56.3	157.2	0.0323	2.6421	
11-15 years	98	119.4	70.0	72.2	355.2	0.0237	2.8297	
ABDOMINOPELVIC								
< 1 year		-	-	-	-	0.0841	-	
1-5 years	36	93.5	36.0	61.6	138.3	0.0537	5.0209	
6-10 years	45	134.8	73.1	74.2	478.3	0.0357	4.8123	
11-15 years	84	226.2	72.1	151.5	1200.0	0.0249	5.6323	

that, in general, the $CTDI_{vol}$ and DLP values recorded were lower than the DRLs reported in the other studies, except for patients between 11-15 years of age who had head scans, suggesting the need for protocol optimisation for this age group. Based on these findings, local pediatric DRLs are proposed for these protocols for this facility.

Data Availability Statement

The datasets used to support the findings of this study are protected by the Institutional Review Board of the Teaching Hospital. However, data are available from the corresponding author for researchers who meet the criteria for access to confidential data.

Conflicts of Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

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